

European Aviation MRO Intelligence Platform

Industry	Aerospace & Aviation	Headquarters	France
Operating Region	France, Germany, United Kingdom, Spain, and the Netherlands	Regulatory Context	GDPR · EU AI Act · Sector-specific EU Directives

Business Challenge

A France-based maintenance, repair, and overhaul (MRO) provider servicing 340 aircraft for six European carriers faces:

- Unscheduled maintenance events causing an average of 11 hours of aircraft-on-ground (AOG) time per incident
- Parts inventory spread across 12 hangars with no real-time visibility, causing cannibalisations
- EASA documentation compliance requiring 4,500 engineer-hours annually
- Engine health data from OEMs delivered in proprietary formats with no unified analytics layer
- Knowledge retention risk as 31% of senior engineers approach retirement age

Transformation Objective

Build an AI-powered MRO intelligence platform that predicts maintenance requirements, optimises parts supply, automates EASA documentation, and preserves engineering expertise.

Azure Services

▸ Azure IoT Hub	▸ Azure Machine Learning
▸ Microsoft Fabric	▸ Azure Synapse Analytics
▸ Azure OpenAI	▸ Dynamics 365 Field Service
▸ Microsoft Purview	▸ Power BI
▸ Azure API Management	▸

Expected Outcomes

AOG time reduced from 11 hours to under 3 hours, parts availability at point of need improved by 34%, EASA documentation effort reduced by 55%, first-time-fix rate improved from 71% to 89%.

AI Infusion Point

A remaining useful life model analyses engine telemetry across multiple OEM data formats. A parts demand forecasting AI optimises stock levels across all 12 hangars. A GenAI engineering assistant retrieves maintenance procedures and completes EASA task cards from structured voice input during maintenance operations.